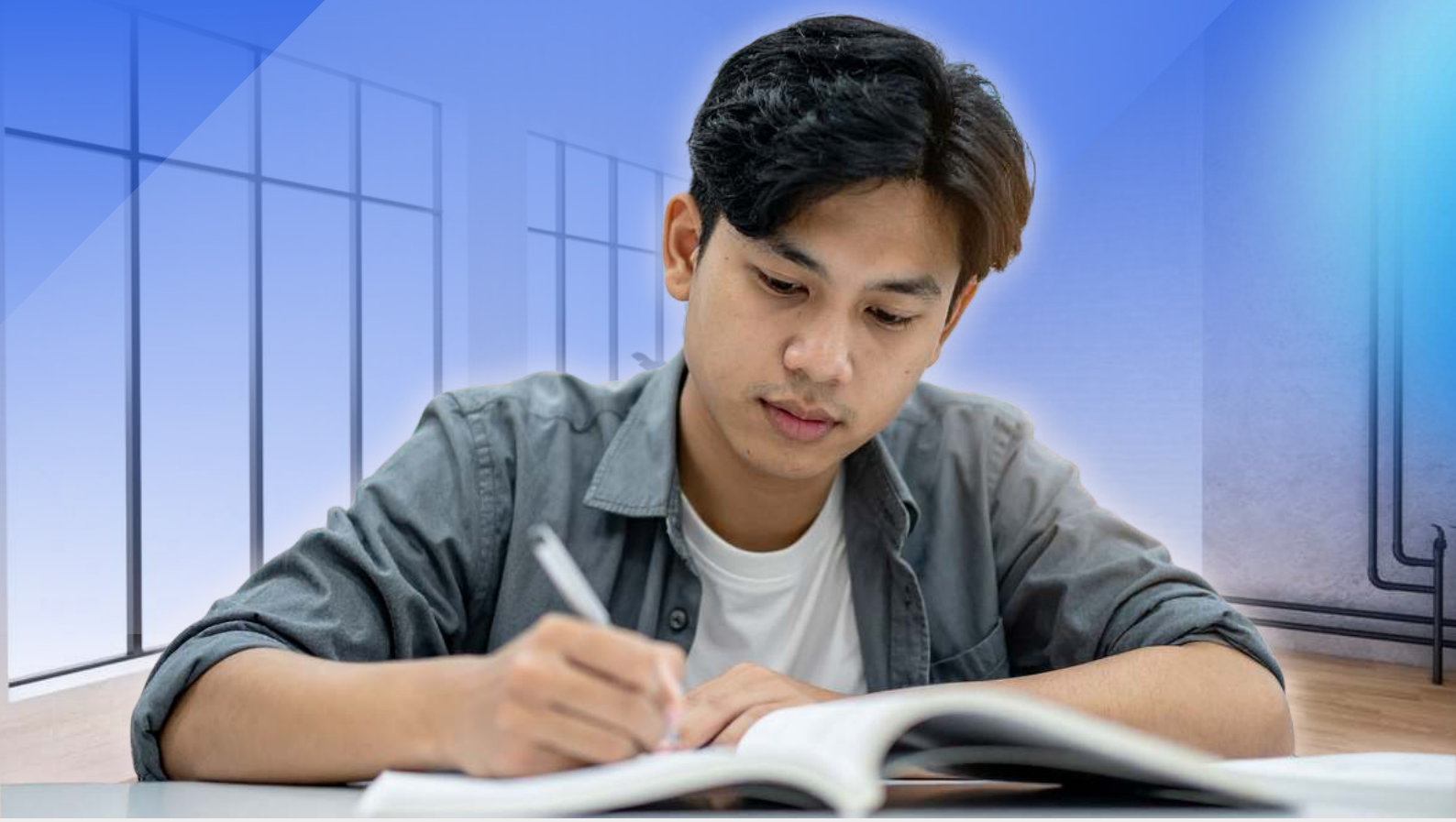
Lenterago

TRY OUT #2

SAT MATH

61003

44 QUESTIONS



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MODUL 1

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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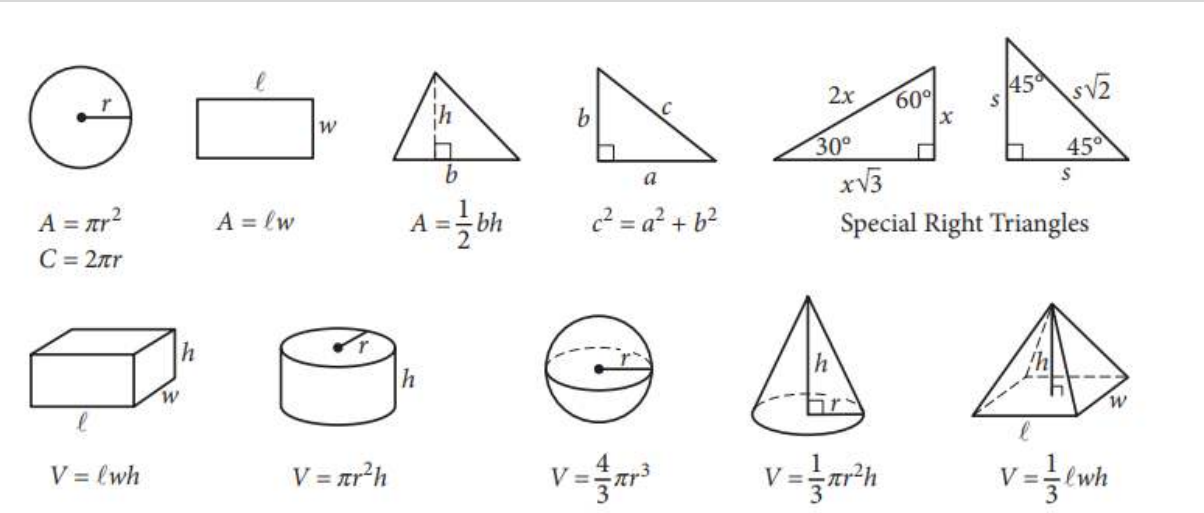

The number of degrees of arc in a circle is 360.

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The sum of the measures in degrees of the angles of a triangle is 180.

0 8 2 1 7 0 2 7 6 0 7 6

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Module 1

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1. Dari pilihan berikut, manakah bentuk aljabar yang setara dengan $\frac{1}{2}x^3\left(x^2-4\right)$?

   A) $\frac{1}{2}x^6-2x^2$

   B) $\frac{1}{2}x^6-2$

   C) $\frac{1}{2}x^5-2x^3$

   D) $\frac{1}{2}x^5-4x^2$

2. On the coordinate plane, a line has slope $\frac{1}{9}$and passes through $(0, 14)$ .Which equation describes this line?

   A) $y=-\frac{1}{9}x-14$

   B) $y=-\frac{1}{9}x+14$

   C) $y=\frac{1}{9}x-14$

   D) $y=\frac{1}{9}x+14$

3. If $\frac{x}{8}=5$,what is $\frac{8}{x}$?

4. Triangles $ABC$and $DEF$are congruent. Vertex Amatches D, and $\angle B$and $\angle E$are right angles. If $m\angle A=18^\circ$,what is $m\angle F$?

   A) 18° B) 72° C) 90° D) 162°

5. Which expression is equivalent to $\left(m^4q^4z^{-1}\right)\left(mq^5z^3\right)$for positive m, q, z?

   A) $m^4q^{20}z^{-3}$

   B) $m^5q^9z^2$

   C) $m^6q^8z^{-1}$

   D) $m^{20}q^{12}z^{-2}$

6. The lines/curves $y=76$ and $y=x^2-5$ meet at some point (x, y) .Which value could $x$be?

   A) $-\frac{76}{5}$

   B) -9

   C) 5

   D) 76

7. A random sample estimates a population proportion as 0.49 with a margin of error 0.04. Which conclusion is most reasonable?

   A) The true proportion could be between 0.45 and 0.53.

   B) The true proportion is likely less than 0.45.

   C) The true proportion equals 0.49.

   D) The true proportion is likely greater than 0.53.

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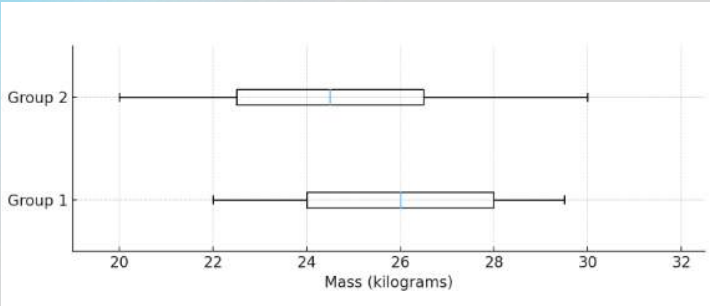
8. 

The box-and-whisker plots show the masses (kg) for two gazelle groups. From these plots, which statement is guaranteed to be true?
A) Group 1's mean mass is greater than Group 2's.
B) Group 1's mean mass is less than Group 2's.
C) Group 1's median mass is greater than Group 2's.
D) Group 1's median mass is less than Group 2's.

9. Two drones fly along routes whose altitudes (in meters) at time $x$minutes are
Drone A: $y = 3x + 12$and Drone B: $y = 3x - 6$.
How many times will the drones be at the same altitude?
A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

10. A water tank is being filled at a constant rate. Its volume is 5liters at $t = 1$hour and 29liters at $t = 7$hours. Which function gives the volume $f(t)$)in liters) after $t$hours?
A) $f(t) = 6t - 1$
B) $f(t) = 4t + 1$
C) $f(t) = t + 4$
D) $f(t) = 4t + 5$

11. A mixture has red:white in the ratio $\frac{x}{y} = 5$ .After multiplying the red amount by 35and the white amount by n, the ratio becomes $\frac{35x}{ny} = 7$ .What is n?

12. A container starts draining through a hole at the bottom. The amount of liquid left after $t$seconds is modeled by
$$w(t) = 300 - 4t \ (milliliters).$$
According to this model, how many milliliters drain out each second?
A) 300 B) 296 C) 75 D) 4

13. An investment grows by 0.49% of its value each year (same percentage increase annually). Which function type best models the value over time?
A) Decreasing exponential
B) Decreasing linear
C) Increasing exponential
D) Increasing linear

14. For the quadratic $y = x^2 - 10x + 13$,at what value of x does y reach its minimum?

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15. Solve the system $10x + y = 50$ and $4x + y = 38$. What is y?

16. Square P has side lengths that are **14 times** the side lengths of square Q. If the area of P is k times the area of Q, what is k?

17. y=−0.80x+28.
When $x = 26.5$, which value is **closest** to the $y$predicted by this model?
A) 6.1
B) 6.8
C) 7.5
D) 8.2

18. Let $h(x) = 4x + 28$ .The graph $y = h(x)$has an x-intercept $(a, 0)$and a y-intercept $(0, b)$ .What is $a + b$?
A) 21 B) 28 C) 32 D) 35

19. Angle R is $\frac{3\pi}{4}$radians. Angle T is $\frac{7\pi}{18}$radians larger than R. What is T in degrees?
A) 185° B) 205° C) 215° D) 305°

20. A culture has 15,000bacteria initially and 60,000after 6hours. If $P = C \cdot 2^{rt}$,what is r?
A) $\frac{1}{6}$ B) $\frac{1}{3}$ C) 6 D) 15,000

21. $\sqrt{(x+1)^2} = \sqrt{4x + 25}$
What is the smallest solution?

22. Given $x - 17 = (x - b)(x - 17)$with $b > 20$. Which are solutions?
I. b
II. $b + 1$ III. 17

A) I & II only
B) I & III only
C) II & III only
D) I, II, & III

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MODUL 2

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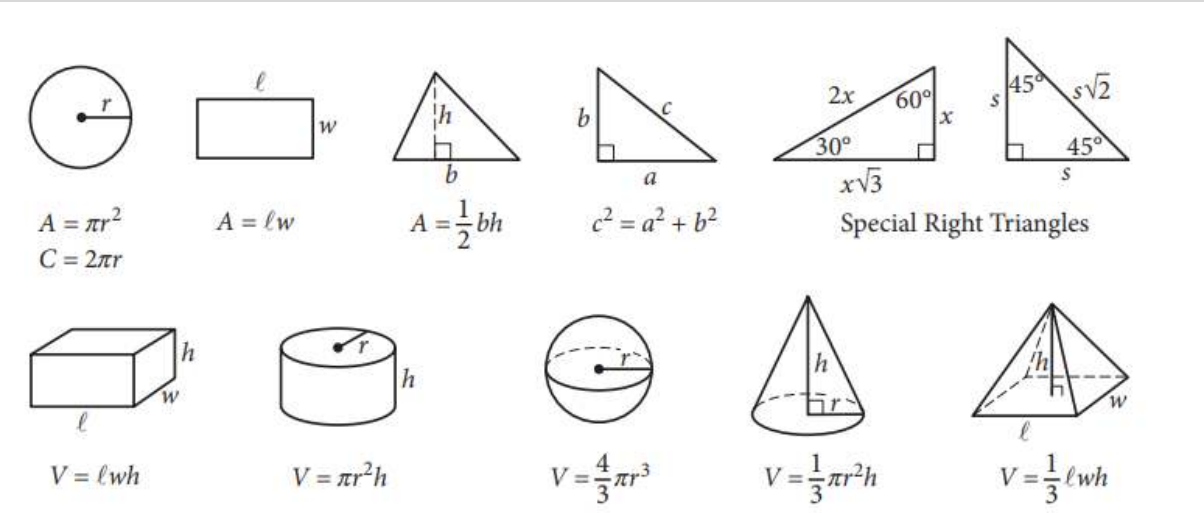

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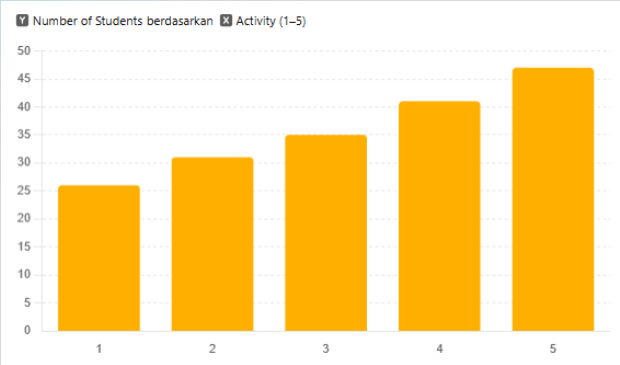
23. What is 15% of 420?
A) 42 B) 63 C) 84 D) 105

24. 

A class voted for five after-school activities. The numbers of students choosing activities 1–5 were 26, 31, 35, 41, 47 respectively. How many students chose activity 3?
A) 28 B) 31 C) 35 D) 41

25. Solve for x:
$5x - 7 = 143.$

26. A shopper spent \$45 on oranges at \$2.50 per pound. How many pounds were purchased?

27. The function f is defined by $f(x) = 7x$ .For what value of $x$does $f(x) = 63$?
A) 6 B) 7 C) 9 D) 63

28. Let $g(x) = x^2 + 4$ .For which value of $x$is $g(x) = 20$?
A) 4 B) 6 C) 8 D) 10

29. A canoe rental charges \$14 per day plus a one-time insurance fee of \$6. Which equation gives the total cost c(in dollars) for renting the canoe for $d$days?
A) $c = 14(d + 6)$
B) $c = 6d + 14$
C) $c = 14d + 6$
D) $c = 20d$

30. Which expression is equivalent to

$$\frac{5}{3x+2} - \frac{2}{x-4}\ ?$$

A) $\frac{1}{(3x+2)(x-4)}$
B) $\frac{x-24}{(3x+2)(x-4)}$
C) $-\frac{x+24}{(3x+2)(x-4)}$
D) $\frac{x+24}{(3x+2)(x-4)}$

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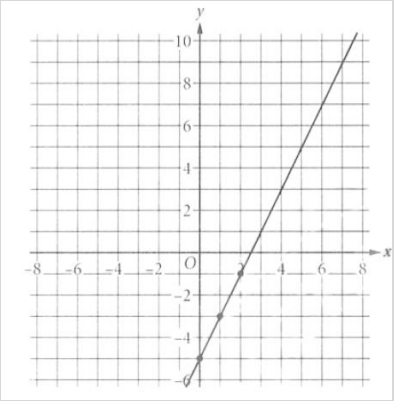
31. Let $f(x) = 7 + 2^x$ .Define $g(x) = 3f(x-2) - 10$.
What is the y-intercept of g?
A) $\left(0, \frac{23}{4}\right)$
B) $\left(0, \frac{47}{4}\right)$
C) $(0, 7)$
D) $(0, 13)$

32. 

The graph shows a linear relationship between $x$and y.
Which table lists three values of $x$and their corresponding y-values for this relationship? (Collage Board, 2023)

a)

x	y
0	0
1	−7
2	−9

b)

x	y
0	0
1	−3
2	−1

c)

x	y
0	−5
1	−7
2	−9

d)

x	y
0	−5
1	−3
2	−1

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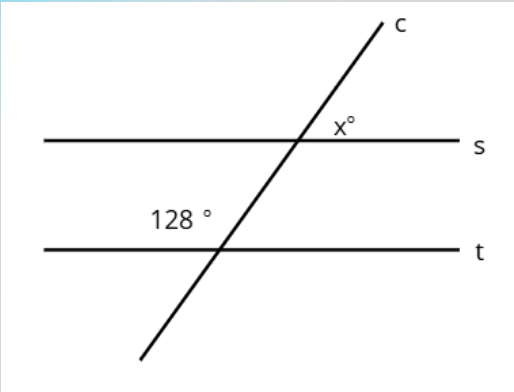
33. 

A transversal p cuts two parallel lines r and s. At line s, the interior angle on the left of p is 128° .The interior angle on the same side of p at line r is labeled x. What is x?

34. A rectangle has width $w$inches and length $L = 2w + 3$inches. If $w = 4.5$, what is the perimeter?
A) 27 B) 30 C) 33 D) 36

35. For distinct positive numbers j, k, m, the relation

$$5j - 2k = 3m$$

holds. Which equation expresses $j$in terms of $k$and m?
A) $j = \frac{3m-2k}{5}$
B) $j = \frac{2k-3m}{5}$
C) $j = 5(3m + 2k)$
D) $j = \frac{3m+2k}{5}$

36. The point $(5, -2)$is on the xy-plane. Which system of inequalities does it satisfy?
A) $x > 0, y > 0$
B) $x > 0, y < 0$
C) $x < 0, y < 0$
D) $x < 0, y > 0$

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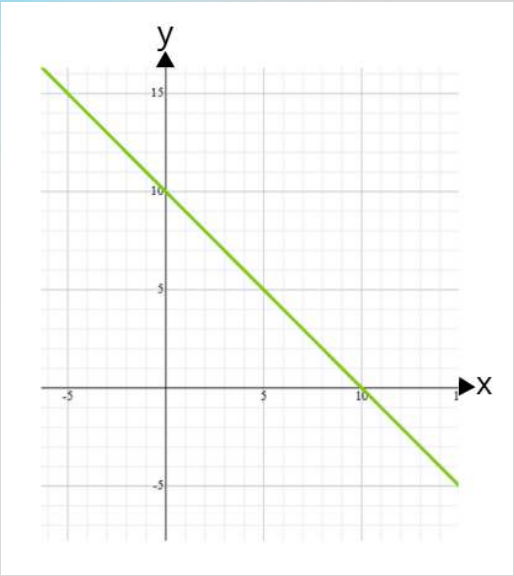
37. 

Based on the graph (downward trend with a positive y-intercept), which equation is the most appropriate linear model?
A) $y = -1.1x - 8$
B) $y = -1.0x + 10$
C) $y = 1.0x - 10$
D) $y = 1.1x + 8$

38. An account is opened with \$50,000. No additional deposits/withdrawals are made. After 8 years the balance is about \$73,872. Which function could model the balance $A(t)$,in dollars, $t$years after opening?
A) $A(t) = 50{,}000(1.05)^t$
B) $A(t) = 73{,}872(1.05)^t$
C) $A(t) = 50{,}000(0.05)^t$
D) $A(t) = 73{,}872(0.05)^t$

39. Solve for the positive solution:
$$5\,|\,x+2\,|+|\,x+2\,|=18$$

40. Find the constant $r$so that
$84y^6 - 56y^5 = r\,y^5(12y - 8)$.

41. The area A(in cm²) of a rectangular cutting board is given by
$$A = w(w + 12),$$
where $w$is the width (cm). Which expression represents the length (cm) of the board?
A) $w(w + 12)$
B) w
C) 12
D) $w + 12$

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42. Consider the inequality $y > 13x - 18$. Which table lists three (x, y)pairs that all satisfy the inequality?
A) $(3, 21), (5, 47), (8, 86)$
B) $(3, 26), (5, 42), (8, 86)$
C) $(3, 16), (5, 42), (8, 81)$
D) $(3, 26), (5, 52), (8, 91)$

43. Let $f(x) = (x-4)(x+1)(x+3)$ .Define $g(x) = f(x+2)$.
The graph of $y = g(x)$has x-intercepts at $(a, 0), (b, 0), (c, 0)$,where $a, b, c$are distinct constants. What is $a + b + c$?
A) -12
B) -6
C) 0
D) 6

44. A square is inscribed in a circle with radius $9\sqrt{2}$inches. What is the side length of the square?
A) 9
B) 12
C) 18
D) $18\sqrt{2}$

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